

Learning Optimal Personalised Reservation Prices in Impression Ad Auctions with Mixture Density Networks

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Abstract

Reservation prices have proven effective in boosting revenue in Generalised Second Price (GSP) auctions, particularly in cost-per-click (CPC) settings. However, in domains like music streaming, where ads are consumed passively without user clicks, a cost-per-impression (CPM) model is more appropriate. Additionally, in the music streaming domain, user intent is typically unknown, unlike in sponsored search, making it essential to optimally leverage all available user and contextual information when setting prices. This paper addresses the challenge of optimising reservation prices in GSP auctions with CPM pricing, adopting a personalised approach that accounts for both user- and advertiser-specific factors.

Using dataset of 100,000 auctions from a major music streaming service, we determine such optimal prices. To achieve this, we first derive the symmetric Nash equilibrium for GSP auctions in a CPM context. We then introduce a Deep Neural Network-based mixture density model that incorporates this equilibrium into its loss. This model captures advertisers' diverse preferences by learning directly from bidding data. We show how this approach enables the computation of personalised prices for both users and advertisers, boosting auction revenue by an average of +4% across ten markets. Our study further highlights the impact of market competitiveness and advertiser preference heterogeneity on revenue gains, showing that personalised pricing greatly enhances auction performance.

CCS Concepts

• **Applied computing** → **Online auctions**; • **Mathematics of computing** → *Maximum likelihood estimation*; • **Computing methodologies** → *Learning latent representations*.

Keywords

Ad Auctions; Revenue Optimisation; Deep Mixture Density Networks; Preference Learning

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1 Introduction

In today's digital landscape, in-session advertisement represents a significant revenue stream for music streaming platforms such as Deezer, Apple Music and Spotify, second only to subscription revenues. Similar to the sponsored search domain [1, 2], a substantial portion of ad inventory on these platforms is allocated through Generalised Second Price (GSP) auctions. These auctions allocate multiple consecutive ad slots within users' listening sessions, allowing advertisers to bid for exposure. However, unlike sponsored search, music streaming is primarily a background experience, meaning users consume in-session ads passively, without clicking on the associated links. As a result, these GSP auctions typically use a cost-per-impression (CPM) rather than a cost-per-click (CPC) pricing model, a format we refer to as *impression GSP*.

In this paper, we leverage a bidding dataset of 100K impression GSP auctions from a major music streaming service to optimise the auction revenue via *personalised reservation prices*. These prices define the minimum bids advertisers must submit to participate in the auction. Setting such prices based on bidding data alone is challenging as the advertisers can strategically adjust their bids in response to the price changes, creating a feedback loop between the price-setting mechanism and the bid-generating process. Therefore, traditional techniques that use demand curves are often ineffective [3–6], making game-theoretic approaches necessary [1, 2, 7].

Personalisation in this setting occurs at both the bidder- (e.g., different prices for video vs. audio ads) and the user-levels (e.g., different prices for different users). To compute such prices, we propose a Deep Neural Network (DNN)-based mixture density network [8] that learns the advertisers' preferences directly from bidding data. To this end, we employ game-theoretic tools to integrate the strategic bidders behaviour directly into the model, breaking the aforementioned feedback loop. This allows us to reconstruct true advertisers' preferences and to estimate the auction revenue in equilibrium for any set of prices. The optimal prices are then obtained by maximising this revenue via gradient descent.

We proceed in three steps. First, we show that the existing methods for setting prices in CPC auctions do not directly apply to impression GSP. Specifically, we demonstrate that the symmetric *Nash equilibrium* (NE) of impression GSP differs from that of CPC auctions [1, 7], highlighting distinct bidder behaviour. We derive the symmetric NE for impression GSP connecting the advertisers' unobservable *values* – their true willingness to pay – with their observable bids.

Second, unlike prior work that focuses on uniform prices, we explore a more general approach where the prices vary across users. Intuitively, advertisers often have *heterogeneous* preferences and may bid more for certain user segments, such as younger users in certain locations. This suggests that *personalised* prices, tailored to the individual user segments, could increase auction revenue. To capture this, we introduce a DNN-based mixture density network that learns advertisers' unobservable values directly from bidding data by incorporating the derived Nash equilibrium into its loss.

Third, we address the challenge of *bidder asymmetry* [2, 9, 10], where the advertisers may be of distinct types such as audio-only vs. video bidders, each with different value distributions. In such cases, traditional methods to compute prices relying only on virtual valuations may not be optimal [2]. Therefore, we propose a Monte Carlo-based approach to estimate the expected revenue in equilibrium for various price configurations. This enables the computation of optimal prices tailored to the different advertiser groups [11].

Our key contributions include:

- **Novel Network Model:** We introduce a DNN-based mixture density network (MDN) that learns the advertisers' heterogeneous preferences and reconstructs their true valuations. To this end, we derive Nash equilibrium of the impression GSP auction and we demonstrate how to incorporate this Nash equilibrium directly into the loss function of the model.
- **Two-Stage Estimation Procedure:** We derive a two-stage parameter estimation procedure that combines maximum likelihood estimation with profile likelihood technique. This allows us to obtain reliable estimates of the discount factor that affects the equilibrium bids when clicking data is unavailable (which is a particular challenge of the impression GSP domain as opposed to the CPC auctions).
- **Revenue Increase & Market Benefits:** Our approach to computing personalised prices significantly boosts auction revenue: Even with a small number of user covariates, we find that personalised prices increase expected revenue by an average of +4% relative to uniform pricing, with gains dependent on the market competitiveness and advertiser heterogeneity.

2 Related Work

In Internet applications, auction mechanisms play a fundamental role in ad allocation. Edelman et al. [12] studied the strategic behaviour of bidders in Yahoo and Google sponsored search first-price auctions. They found that bidders exhibited unstable bidding behaviour, aligning with economic theory, and recommended modifying auction rules to achieve more stable outcomes. However, our focus is on studying GSP auctions rather than first-price auctions.

Edelman et al. [1] and Varian [7] analysed the Nash equilibrium of the widely used GSP auction format for allocating multiple ad

slots. They demonstrated that GSP auctions are not truthful and derived the symmetric NE for GSP with CPC pricing. However, as we show in Section 3.3, the utility models in CPM and CPC pricing differ, resulting in different equilibria. Consequently, these findings do not directly apply to our domain of impression GSP auctions.

In his seminal work, Myerson [13] introduced the design of the revenue-optimal single-item auction, proving that the Vickrey-Clarke-Groves (VCG) mechanism [14–16] with reservation prices is optimal in symmetric domains where bidders draw their values from the same distribution. He also explored the asymmetric domain, where bidders have values drawn from different distributions. Building on this and earlier studies [1, 7], Ostrovsky et al. [2] optimised revenue for Yahoo sponsored search GSP auctions in the symmetric case, demonstrating through a field experiment that applying Myerson reservation prices [13] increases the auction revenue. Additionally, Gomez et al. [17] studied the Bayesian-Nash equilibrium of GSP under CPC pricing. While our approach shares similarities with these works, we focus on CPM rather than CPC pricing and aim to compute personalised prices instead of uniform ones [2], addressing a more general asymmetric case.

Revenue optimisation in sponsored search has also been explored by Lahaie et al. [9], who conducted extensive simulations of ranking rules to examine how the correlation between advertisers' values and their click-through rates (CTRs) impacts auction revenue. Their work focused on bid reweighting rather than reservation price computation. Jeunen et al. [10] conducted simulation-based studies of single-item second-price auctions, and Derakhshan et al. [18] investigated personalised reservation prices in second-price auctions. Unlike these studies, we focus on the more commonly used GSP auction format in the ads domain. Finally, Tang et al. [19, 20] explored RL techniques for computing reservation prices. Contrary to this work, we ground our work in microeconomic theory to allow for better attribution of the revenue gains and explainability.

3 Auction Design

We begin by introducing the concept of impression ad auctions and presenting a formal model of impression GSP. We then illustrate that similarly to CPC auctions, the advertisers in impression GSP do not bid truthfully. To this end, we derive the NE strategies of the advertisers that link the advertisers' private values (unobservable by the auctioneer) to their observable bids. Using this equilibrium, we show how the changes in reservation prices affect the bidding behaviour of the advertisers, and we propose a gradient descent method to set prices, optimising auction revenue in equilibrium.

This section outlines our theoretical framework for calculating personalised reservation prices, assuming the auctioneer has knowledge of the relevant value distributions. In Section 4, we address the main challenge of learning these value distributions from bidding data by presenting our DNN-based mixture density network, that incorporates the derived Nash equilibrium into its loss function.

3.1 In-Session Ads Auction

We begin by describing the in-session ads domain, as shown in Figure 1. In this setting, a user starts a listening session by streaming music tracks from one of the available playlists on a music streaming service. Periodically (e.g., every 15 minutes), the app generates an *ad request* (e.g., “Ad request 1” in Figure 1).

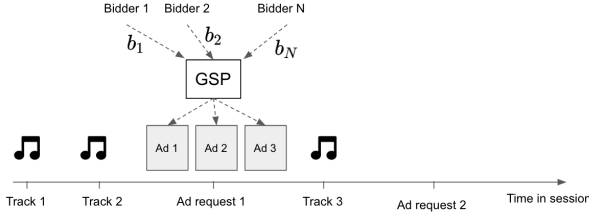


Figure 1: Illustration of in-session ads.

When an ad request is generated, a GSP auction is instantiated to allocate $K \in \mathbb{N}$ consecutive ad slots. Advertisers bid for these slots, targeting specific user characteristics such as age, gender, and market. The auction then determines the winning ads and charges the corresponding bidders. Once the ads are allocated, the user must listen to all K ads consecutively, with no option to skip them. After the ads play, the user resumes their session until the next request arrives (“Ad request 2” in Figure 1). At that point, a new GSP auction is instantiated, and the process repeats.

We assume that auctions corresponding to different ad requests are independent, and focus on optimising the revenue of a specific auction instance.

3.2 Notation & Formal Auction Model

We let $y \in \mathbb{R}^s$ represent the vector of user covariates (e.g., age, gender, country), where $s \geq 0$ is the number of available covariates, and let $N \in \mathbb{N}$ be the number of bidders in the auction. We define $\theta_j \in \{0, 1\}$ as the *type* of bidder j , where $\theta_j = 1$ if bidder j is a *video bidder* and $\theta_j = 0$ if bidder j is an *audio bidder*, for $j = 1, \dots, N$.¹ Let $\theta = (\theta_1, \dots, \theta_N)$ represent the *type profile* of all bidders.

Values vs. Bids. Let $v_j \in \mathbb{R}_{\geq 0}$ be the private *value* of bidder j for displaying their ad to a user, and let $v = (v_1, \dots, v_N)$ denote the *value profile* of all bidders. We assume that value v_j of bidder j , with type θ_j , for user y is independently drawn from density $f_{\theta_j}(v|y)$. This reflects that the audio and video bidders may have different value distributions, and have heterogeneous preferences over users y .

We assume that the earlier slots in the auction are more valuable to the bidders than the later ones. In the music streaming domain, users may pay less attention to ads in the later slots compared to the earlier ones, for example, by lowering the volume of their app or multitasking. Consequently, the value of bidder j for slot $k = 1, \dots, K$ is $v_j \delta^k$, where $\delta \in (0, 1)$ is the *discount factor*. We assume that all bidders know δ , and it is the same for everyone. Finally, we assume that bidders have zero value if they do not get a slot.

Bidder Behaviour and Value Reconstruction. Let $b_j \in \mathbb{R}_{\geq 0}$ represent the *bid* submitted by bidder j , and let $b = (b_1, \dots, b_N)$ denote the *bid profile* of all bidders. Bidders are assumed to be *strategic*, meaning their bids b do not necessarily match their true values v [21]. Importantly, the auctioneer can observe the reported bids b and the types θ , but not the actual values v of the bidders.

In Section 3.4, we discuss that to set prices requires the auctioneer to know how much advertisers are willing to pay (i.e., their value distributions $f_{\theta_j}(v|y)$). However, since these values are unobservable, the auctioneer’s main challenge is inferring them from the observable bids. This process is detailed in Sections 3.3 and 4.

¹Generally, our approach can be used with any finite number of bidder types.

Allocation, Payments and Utility. Let p_j be the payment made by bidder j in the auction, and let $a_j \in \{1, \dots, K\}$ represent the slot allocated to bidder j , with $a_j = \infty$ if no slot is allocated. The *quasi-linear utility function* $u_j : \{1, \dots, K, \infty\} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ of bidder j with value v_j , is defined as follows:

$$u_j(a_j, p_j | v_j) = v_j \delta^{a_j} - p_j. \quad (1)$$

This utility reflects that bidders prefer to secure the most valuable slot while minimizing their payment. Notably, this differs from the commonly used utility function in the CPC domain, which is defined as $u_j^{CPC}(a_j, p_j | v_j) = \delta^{a_j}(v_j - p_j)$ [1, 7].

We let $r(\theta_j, y) \in \mathbb{R}_{\geq 0}$ be the *reservation price* set by the auctioneer for bidder j with type θ_j , when bidding for user y . If bidder j bids less than $r(\theta_j, y)$, they will not be allocated a slot (i.e., only bidders with sufficiently high bids can participate in the auction). The goal of our paper is to determine how to set these prices for different users y and bidders θ_j to maximise the auction revenue.

GSP Auctions with CPM Pricing. In our setting, impression GSP ranks bidders by their bids. Without loss of generality, we assume that $b_1 \geq b_2 \geq \dots \geq b_N$. The bidder with the highest bid b_1 is allocated the first slot, the second-highest bid b_2 gets the second slot, and so on. Ties are broken randomly.

The payment function $p(b_1, \dots, b_N) = (p_1, \dots, p_N)$ is defined as:

$$p_j = \begin{cases} b_{j+1} & \text{if } j \leq K \\ p_j = 0 & \text{if } j > K. \end{cases} \quad (2)$$

If a bidder is allocated to slot j , they pay the $(j + 1)$ ’th highest bid. Non-allocated bidders pay nothing. In this setting, users cannot skip ads, and payments are charged as soon as the ads start playing. Finally, we rely on the standard definition [21] of Nash equilibrium.²

Other Notation. Let X be a random variable with the density $f(X|\mu, \sigma)$, where μ and σ are the parameters of the density function. We denote $f_{(k, \ell)}(X|\mu, \sigma)$ as the density of the k -order statistic [22] from a sample of ℓ i.i.d. variables $X_1, \dots, X_\ell$ drawn from $f(\cdot|\mu, \sigma)$.

With this notation in place, we next demonstrate how to connect the unobservable values v of advertisers with their observable bids b through Nash equilibrium.

3.3 Nash Equilibrium in Impression GSP

We now illustrate the derivation of a symmetric Nash equilibrium (NE) for the impression GSP, emphasizing how this equilibrium differs from the one commonly used in the CPC pricing model [2].

Let us define the bids profile $b^* = (b_1^*, \dots, b_N^*)$ of N bidders recursively as follows:

$$b_j^* = \begin{cases} (\delta^{j-1} - \delta^j)v_j + b_{j+1}^* & j = 1, \dots, K \\ \delta^{j-1}v_j & \text{otherwise.} \end{cases} \quad (3)$$

The following theorem establishes that the bids b^* defined in Equation 3 constitute a Nash equilibrium in the impression GSP:

Theorem 1. b^* is a Nash equilibrium of the impression GSP auction.

PROOF. The proof of this theorem is provided in Appendix A.2. $\square$

²For completeness, we duplicate all definitions in Appendix A.1.

In our paper, we refer to equilibrium b^* , as defined in Equation 3, as the *symmetric NE* of the impression GSP auction. The prior work of Edelman et. al. [1] has proven that such NE arises as the unique perfect Bayesian equilibrium of the English auction, mimicking the bidding dynamics observed in practical ad auctions. Additionally, [2] has shown empirically that relying on such symmetric NE helps to improve the auction revenue in the real-world settings. Thus, we choose to focus on this specific solution concept in our work.

While the bidders' values v allow for the direct computation of their equilibrium bids via Eq. 3, these values remain unobservable to the auctioneer, who needs them to set the prices. This presents the auctioneer with an inverse problem: determining the density $f_{\theta_j}(v|y)$ of the unobservable values that are *most likely* to have generated the observable bids b^* . In Section 4, we introduce our model for estimating this density, while for the remainder of this section we assume that the auctioneer has already learned $f_{\theta_j}(v|y)$.

Impact of Reservation Prices. The prices can affect the NE and are typically incorporated into the auction via the auctioneer's bid [13]. A positive price $r(\theta_j, y) > 0$ can have two opposing effects:

- (1) **Incentivizing higher bids:** If the reservation price is sufficiently high (i.e., $r(\theta_K, y) > \max\{b_{K+1}^*, \dots, b_N^*\}$), the auctioneer effectively acts as the $(K+1)$ 'th highest bidder. As a result, the K 'th bidder increases their bid from $(\delta^{K-1} - \delta^K)v_K + b_{K+1}^*$ to $(\delta^{K-1} - \delta^K)v_K + r(\theta_K, y)$. Due to the recursive structure of Equation 3, the rest $K-1$ allocated bidders will also increase their bids, leading to higher auction revenue.
- (2) **Cancelling allocations:** If the reservation price $r(\theta_j, y)$ is set too high, many potential allocations may be cancelled, leading to no payments being made to the auctioneer. This, in turn, reduces the auctioneer's revenue.

We now demonstrate how to compute the optimal prices in impression GSP auctions to maximise the auctioneer's revenue, assuming bidders adhere to their symmetric Nash equilibrium strategies.

3.4 Optimal Reservation Prices

We illustrate how to compute optimal prices in the most general asymmetric case, where bidders of different types draw values from different distributions. We begin by presenting an algorithm to calculate the auctioneer's revenue for any given reservation prices, and we then use this algorithm, in combination with gradient descent, to find the prices that maximise the auction revenue.

For now, we assume the densities $f_{\theta}(v|y)$ are known, allowing us to sample bidders' values from these densities (In Section 4, we describe how to estimate $f_{\theta}(v|y)$ from bidding data). Let $M \in \mathbb{N}$ denote the number of Monte Carlo samples used to estimate the expected auction revenue. Algorithm 1 outlines our approach.

In line 3, we sample the value v and type θ profiles of bidders, and then sort v in non-increasing order (line 4). We only retain bidders whose expected values $\delta^j v_j$ exceed their reservation prices $r(\theta_j, y)$ (line 5). Let N denote the number of such bidders. If N exceeds the number of slots K , we recompute the equilibrium strategies (lines 7-9) according to Equation 3. Here, we see that if the reservation price of the last allocated bidder is sufficiently high (line 8), all allocated bidders increase their bids (line 9). Lines 11-12 handle cases when $N < K$. Finally, we compute the auction revenue as the total GSP payment from the allocated bidders (line 14).

```

Input:  $M \geq 0, f_{\theta}(v|y), \delta, r(\theta, y), y$ 
Result:  $Revenue \leftarrow rev$ 
1  $rev \leftarrow 0;$ 
2 while  $i \leq M$  do
3    $(v, \theta) \leftarrow \text{sample\_value\_profile}(f_{\theta}(v|y))$ 
4    $v \leftarrow \text{sort\_desc}(v)$ 
5    $v \leftarrow \{v_j : \delta^j v_j \geq r(\theta_j, y)\}, \quad N \leftarrow |v|$ 
6   if  $N \geq K$  then // Compute Nash eq.
7      $b_j^* \leftarrow \delta^{j-1} v_j \quad \forall j = K+1, \dots, N+1$ 
8      $b_{K+1}^* \leftarrow \max\{b_{K+1}^*, r(\theta_K, y)\}$ 
9      $b_j^* \leftarrow (\delta^{j-1} - \delta^j) v_j + b_{j+1}^* \quad \forall j = 1, \dots, K$ 
10  else
11     $b_{N+1}^* \leftarrow r(\theta_N, y)$ 
12     $b_j^* \leftarrow (\delta^{j-1} - \delta^j) v_j + b_{j+1}^* \quad \forall j = 1, \dots, N$ 
13  end
14   $rev \leftarrow rev + \sum_{j=1}^{\min\{K, N\}} b_{j+1}^*$ 
15   $i \leftarrow i + 1$ 
16 end
17  $rev \leftarrow \frac{rev}{M}$ 

```

Algorithm 1: Estimation of the auction revenue.

Algorithm 1 enables us to compute the equilibrium revenue for any set of prices $r(\theta, y)$. To determine the optimal prices $r^*(\theta, y)$ for any user y and bidder type θ_j , we apply numerical gradient descent to maximise the auction revenue estimated by this algorithm.

4 Learning Advertisers' Preferences

In this section, we introduce a mixture density network (MDN) to estimate the unobservable densities $f_{\theta}(v|y)$, needed to compute optimal prices (Section 3.4). We begin by describing the data context and formally defining the probabilistic model of the bid-generating process. We then derive the likelihood of observing bids under this probabilistic model in the simpler case of *homogeneous* bidder preferences (ignoring user covariates y). Finally, we show how this likelihood can serve as a loss function for our MDN in the more general case of *heterogeneous* bidder preferences.

4.1 Data Context

We have an access to a dataset $\mathcal{D}$ containing M auction instances. For clarity, we use superscripts to denote the specific auction instances. Let N^i represent the number of bidders in the i 'th auction instance, where $i = 1, \dots, M$. Similarly, let $v^i = (v_1^i, \dots, v_{N^i}^i)$ represent the value profile of the bidders in that auction instance, and let b^i, θ^i and y^i denote the respective bid profiles, type profiles, and user covariates. W.l.o.g., we assume that bids are ordered in non-increasing order, i.e., $b_j^i \geq b_k^i$ for any $k \geq j$ and for all $i = 1, \dots, M$. Therefore, we can define our dataset as $\mathcal{D} = \{(b^i, \theta^i, y^i)\}_{i=1}^M$.

4.2 Likelihood in the Homogeneous Case

Let N^i be drawn i.i.d. from the beta-binomial distribution, i.e., $N^i \sim \text{BetaBin}(\cdot | n, \alpha, \beta)$, where $n, \alpha, \beta \geq 0$ are unknown scalar parameters. Additionally, we assume that bidders' types are drawn i.i.d. from Bernoulli distribution, i.e., $\theta_j^i \sim \text{Ber}(\pi)$, where $\pi \in [0, 1]$. Thus, the values v_j^i are drawn from a mixture model $\tilde{f}(v|\mu_0, \sigma_0, \mu_1, \sigma_1, \pi)$:

$$\tilde{f}(v|\mu_0, \sigma_0, \mu_1, \sigma_1, \pi) = \pi f(v|\mu_0, \sigma_0) + (1 - \pi) f(v|\mu_1, \sigma_1),$$

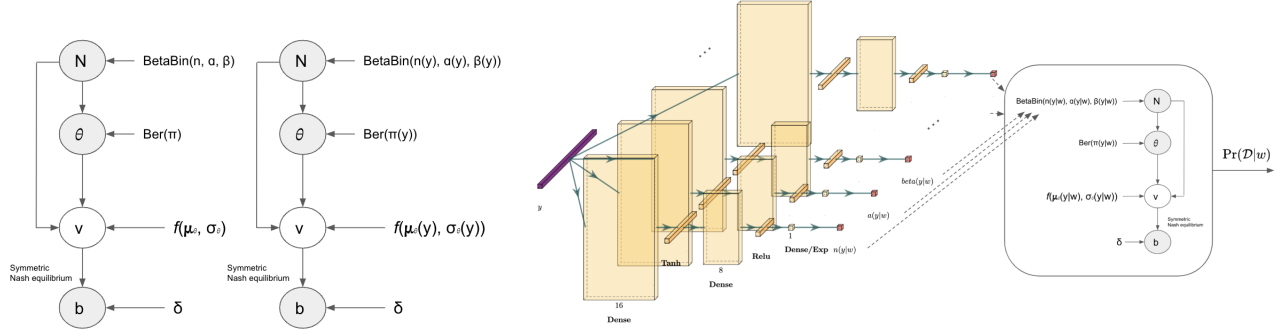


Figure 2: Learning advertisers' value distributions. Left: Probabilistic bid-generating models for (1) homogeneous and (2) heterogeneous advertisers' preferences. The shaded and the unshaded circles correspond to the observable and the unobservable quantities, respectively. Right: Mixture density network for learning advertisers' heterogeneous preferences. The yellow boxes correspond to the layers and activation functions of respective feedforward DNNs.

where $f(v|\mu_0, \sigma_0)$ corresponds to bidders of type $\theta_j^i = 0$, and $f(v|\mu_1, \sigma_1)$ corresponds to bidders of type $\theta_j^i = 1$.³ Thus, the full probabilistic model is illustrated in Figure 2 (left, 1st figure), with the set of parameters to be estimated $\{n, \alpha, \beta, \mu_0, \sigma_0, \mu_1, \sigma_1, \pi, \delta\}$.

A particular challenge of impression GSP auctions as opposed to CPC auctions is that some of these parameters may not be identifiable: Indeed, the small observed bids b could arise either from bidders having small values v , or from bidders having large values v but a small discount factor δ (see Equation 3). To address this challenge, we propose a two-stage estimation procedure:

- (1) **Maximum Likelihood Estimation (MLE):** First, we fix the discount δ to a certain value, and we rely on MLE to identify the rest of the parameters $n, \alpha, \beta, \mu_0, \sigma_0, \mu_1, \sigma_1$ and π .
- (2) **Profile Likelihood:** We repeat step (1) for a discretized interval of discounts δ , and use profile likelihood [23] to identify the optimal δ_{OPT} . We then construct the confidence interval for δ , and we leverage the upper- and lower- confidence bounds on δ to estimate the value distribution.

In the rest of this section, we elaborate on these two steps.

Maximum Likelihood Estimation. The likelihood of observing dataset $\mathcal{D}$ given the model parameters can be expressed as follows:

$$\Pr(\mathcal{D}) = \prod_{i=1}^M \text{BetaBin}(N^i | n, \alpha, \beta) \times \prod_{j=1}^{\min(N^i, K)-1} \tilde{f}_{N^i-j+1, N^i} \left(\frac{b_j^i - \delta b_{j+1}^i}{1 - \delta} \middle| \mu_0, \sigma_0, \mu_1, \sigma_1, \pi \right) \times \prod_{j=1}^{\max(N^i - K, 0)+1} \tilde{f}_{N^i-(K+j)+1, N^i} \left(\frac{b_{K+j}^i}{\delta^{K+j-1}} \middle| \mu_0, \sigma_0, \mu_1, \sigma_1, \pi \right) \times \left[\tilde{f}_{1, N^i} \left(\frac{b_{N^i}^i}{1 - \delta} \middle| \mu_0, \sigma_0, \mu_1, \sigma_1, \pi \right) \right]^{1_{\{N^i < K\}}}.$$

Here, the first term corresponds to the probability of observing exactly N^i bidders in the i 'th auction instance. Recall from Lemma 1 that the equilibrium bids are value-ordered. Thus, the value v_j^i of the

j 'th highest bidder in auction instance i must follow the distribution of the $(N^i - j + 1)$ 'th order statistic from a sample size N^i drawn from $\tilde{f}$. We can now rely on the Nash equilibrium from Theorem 1 to connect the unobservable values with the observable bids.

In particular, for $j = 1, \dots, \min(N^i, K)$, the value v_j^i of the j 'th highest bidder is $v_j^i = \frac{b_j^i - \delta b_{j+1}^i}{1 - \delta}$ (see Eq. 3). Thus, the respective value density is $\tilde{f}_{N^i-j+1, N^i} \left(\frac{b_j^i - \delta b_{j+1}^i}{1 - \delta} \middle| \cdot \right)$. This corresponds to the second term in Eq. 4. Similarly, the third term in Eq. 4 accounts for the unallocated bidders ($j > K$). If the number of bidders N^i is smaller than K , the value of the lowest bidder is $v_j^{(i)} = \frac{b_j^i}{1 - \delta}$. This forms the final term in Eq. 4. We can now use stochastic gradient descent [24] to estimate the parameters that maximise log-likelihood $\log \Pr(\mathcal{D})$.

Profile Likelihood. We discretise the interval $(0, 1)$ and perform MLE estimation for all discounts δ within the respective range. Using the resulting log-likelihood values, we construct the profile likelihood [25] and determine the optimal δ_{OPT} along with the confidence interval $(\delta_{\text{LCB}}, \delta_{\text{UCB}})$. In Section 5, we demonstrate how this enables more robust estimates of the optimal prices and revenues.

Maximising likelihood in Equation 4 allows us to obtain value distributions f_{θ} that are independent of the user y . Next, we illustrate how to learn fully personalised distributions $f_{\theta}(\cdot|y)$ in the most general heterogeneous setting.

4.3 Learning Heterogeneous Preferences

To model bidders' heterogeneous preferences we now define $n, \alpha, \beta, \mu_0, \sigma_0, \mu_1, \sigma_1$ and π as parametric models of the user covariates y , rather than as scalars. Specifically, we let $n = n(y|w_n), \alpha = \alpha(y|w_\alpha), \beta = \beta(y|w_\beta), \mu_0 = \mu_0(y|w_{\mu_0}), \sigma_0 = \sigma_0(y|w_{\sigma_0}), \mu_1 = \mu_1(y|w_{\mu_1}), \sigma_1 = \sigma_1(y|w_{\sigma_1}), \pi = \pi(y|w_\pi)$, where $w_n, w_\alpha, w_\beta, w_{\mu_0}, w_{\mu_1}, w_{\sigma_0}, w_{\sigma_1}, w_\pi \in \mathbb{R}^q$ are the respective parameters in some q -dimensional space (see Figure 2, left, 2nd figure). Each of these parametric models is now represented with a DNN, where the input corresponds to the user covariates y , and the multi-dimensional output corresponds to $n, \alpha, \beta, \mu_0, \sigma_0, \mu_1, \sigma_1, \pi$ (see Figure 2, right). In practice, for each of these models we use 3 dense layers of sizes 16-8-1, relu activation for the hidden layers, and we initialise weights uniformly at random.

To ensure non-negativity of $n, \alpha, \beta, \mu_0, \sigma_0, \mu_1$ and σ_1 (Section 4.2), we use an exponential activation function in the last layer of the network. For the Bernoulli parameter π , which must remain between

³Generally, $f(\cdot)$ can be any density with a non-negative support, e.g., log-normal, Gamma or Beta. In our work, we will rely on Beta densities.

0 and 1, we use a softmax activation. Consequently, the weight matrices of the DNN correspond to the parameters w_n , w_α , w_β , w_{μ_0} , w_{μ_1} , w_{σ_0} , w_{σ_1} and w_π , which are learned from bidding data $\mathcal{D}$.

To learn these weights, we define the loss function as the negative log-likelihood according to Equation 4, where we replace each scalar n , α , β etc. with their parametric counterpart $n(y|w_n)$, $\alpha(y|w_\alpha)$, $\beta(y|w_\beta)$, and so on. As a result, our final model (Figure 2, right) represents a MDN [24] for modelling the bid-generating distribution, incorporating the Nash equilibrium directly into its loss. As in Section 4.2, we rely on stochastic gradient descend and our two-step estimation procedure to identify the unknown parameters of the model.

5 Experiments

We carried out a number of experiments to evaluate our approach, having two main goals: (1) To validate our method using synthetic data where the ground truth value distributions are known, and (2) To compute the optimal personalised prices for a number of existing markets by leveraging a real bidding dataset of 100,000 impression GSP auctions from a major music streaming service.

Baselines. We compare our approach against the following baselines:

- *Zero reservation prices (ZP).* This is a weak baseline that sets all reservation prices to zero.
- *Myerson prices (MP).* We rely on the seminal Myerson paper [13] to compute the optimal uniform reservation prices.
- *Empirical Revenue Maximisation (ERM).* Using the cumulative distribution (CDF) of the observed bids we estimate the demand curve as 1-CDF. We then choose the prices that set the marginal revenue computed with this demand curve to zero (see [3]).
- *Single Sample Mechanism (SSM).* The approach proposed in [26] approximates the Bayesian optimal mechanisms by sampling the reservation prices from the respective value distributions.
- *Personalised Mixture Density Network (P-MDN).* Our proposed approach as introduced in Sections 3-4.

5.1 Synthetic Data

We generate $M = 10K$ auction instances. For each instance i , we first draw the binary type of the user $y \in \{0, 1\}$ from Bernoulli distribution, $y \sim \text{Ber}(0.6)$ (e.g., y could represent the user's gender). Next, we draw the number of bidders from $N^i \sim \text{BetaBin}(20, 3, 6)$ if $y = 0$, or $N^i \sim \text{BetaBin}(10, 5, 4)$ if $y = 1$. The ground truth distributions for these are depicted as thin solid lines in Figure 3 (left). We set the fraction of audio bidders ($\theta = 1$) to $\pi = 0.7$, i.e., $\theta_j^i \sim \text{Ber}(\pi)$ for all $j = 1, \dots, N^i$ and $i = 1, \dots, M$.

The ground truth parameters for the value densities (Beta distributions) are $\mu_0 = 4.4$, $\sigma_0 = 24$, $\mu_1 = 6$, $\sigma_1 = 2$ for users of type $y = 0$, and $\mu_0 = 1.2$, $\sigma_0 = 10$, $\mu_1 = 4$, $\sigma_1 = 3$ for users of type $y = 1$. Figure 3 (center) shows the corresponding ground truth densities (thin solid lines). Finally, we set the discount factor to $\delta = 0.6$. For each auction instance i , we drew the value/type profiles using the probabilistic model in Figure 2 (right) and computed the respective equilibrium bids b^{*i} using Equation 3. The resulting histogram is depicted in blue in Figure 3 (right). Unlike the value distribution (shown in orange), the bid distribution is concentrated closer to

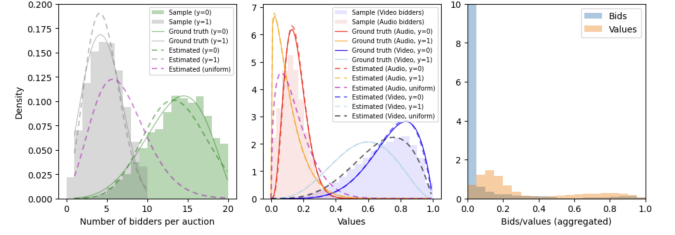


Figure 3: Left: Distribution of the number of bidders per auction. Centre: Densities of the values of bidders of different types. Right: Values and the respective equilibrium bids.

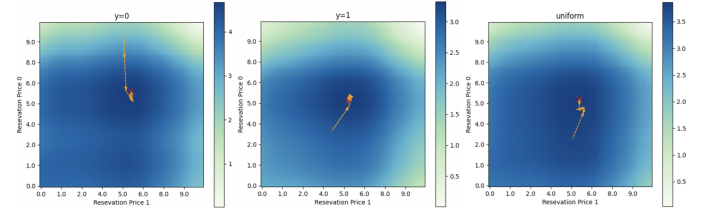


Figure 4: Revenue surface and optimal reservation prices. Left: users of type $y = 0$. Centre: users of type $y = 1$. Right: uniform case.

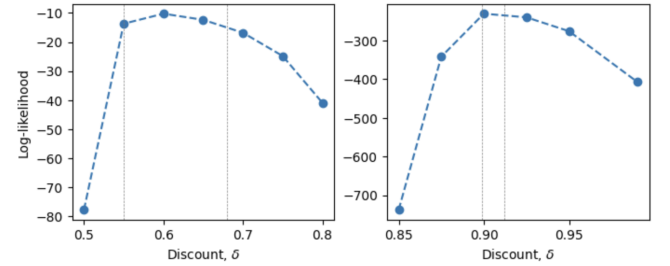


Figure 5: Profile likelihood. Left: Synthetic Data. Right: Real Auction Data. Vertical lines indicate the 95% confidence intervals for the discount factor δ .

zero, which is expected, as bidders lower their bids relative to their values in equilibrium (see Equation 3).

As discussed in Section 4, we perform parameter identification relying on our two-step estimation procedure. First, Figure 5 (left) illustrates the profile likelihood constructed for the different values of δ by maximising the log-likelihood $\log \Pr(\mathcal{D})$ (Eq. 4) while keeping the respective discount δ constant. We see that the value of δ that maximises this likelihood profile matches the ground truth value $\delta = 0.6$. The vertical lines indicate the confidence interval at 0.05 significance level. We now proceed to computation of prices for $\delta_{\text{OPT}} = 0.6$, and we provide the respective summaries for the lower (δ_{LCB}) and upper (δ_{UCB}) confidence bounds in Table 1.

For $\delta_{\text{OPT}} = 0.6$, the dashed green and gray lines in Figure 3 (left) represent the estimated distributions of the number of bidders across different user types. The purple line shows the estimated distribution for the homogeneous case. Similarly, in Figure 3 (center), the dashed red, orange, blue, and light blue lines show the estimated value densities for audio and video bidders across different user types, while purple and black lines represent the estimated densities for the homogeneous case. We see that all the estimated densities closely approximate the ground truth.

Discount	Opt. Prices (vid/aud)		Revenue gains of P-MDN rel. to:			
	y=0	y=1	ZP	MP	ERM	SSM
$\delta_{OPT} = 0.6$	6.38 / 6.25	4.5 / 4.5	+81.0%	+3.4%	+3.3%	+19.9%
$\delta_{LCB} = 0.59$	6.75 / 6.50	4.38 / 4.38	+82.4%	+4.9%	+4.9%	+21.3%
$\delta_{UCB} = 0.65$	5.75 / 5.88	4.88 / 4.62	+69.6%	+2.5%	+1.0%	+13.5%

Table 1: Optimal prices and the respective revenue gains of P-MDN relative to ZP, MP, ERM and SSM baselines.

Using the estimated value distributions, we now compute the optimal prices. To this end, we apply numerical gradient descent and Algorithm 1 to identify the prices that maximise the expected auction revenue in equilibrium. Figure 4 shows the revenue surfaces for users of type $y = 0$ (left) and type $y = 1$ (center), with the steps of the gradient descent highlighted by the orange arrows.⁴ Figure 4 (right) displays the revenue surface for the uniform (homogeneous) case. In all figures, the red crosses mark the optimal prices.

We observe that the optimal prices vary depending on the user type (the specific prices are shown in Table 1). Comparing the optimal revenue in Figure 4 with the revenue at the origin, we see that P-MDN results in a significant +81% revenue increase compared to zero prices (ZP). The large gains here are not surprising as ZP is a weak baseline. Moreover, using personalised prices, instead of the uniform ones (MP), yields another +3.4% revenue gains. We also see that P-MDN significantly overperforms ERM (+3.3%) and SSM (+19.9%). Table 1 provides the respective estimates for the lower (LCB) and upper (UCB) confidence bounds of the discount.

5.2 Real Bidding Data

We conducted a similar experiment using bidding data from 100,000 real impression GSP auctions on a major music streaming platform. Each user’s demographic data y (market and age) was used to compute the personalised prices. For simplicity, we present the analysis for two European markets, referring to them as ‘Country 1’ and ‘Country 2’ for confidentiality, with currency units normalised. We also group users into two age categories. In the following section, we extend our analysis to ten additional real markets.

In our experiment, we define $y = (y_1, y_2) \in \{0, 1\}^2$, where $y_1 = 1$ if the user is from ‘Country 1’ and $y_1 = 0$ if they are from ‘Country 2’. Similarly, $y_2 = 1$ if the user is younger than 20 years old, and $y_2 = 0$ otherwise. The parametric models for $n, \alpha, \beta, \sigma_0, \mu_1, \sigma_1, \pi$ are defined as discussed in Section 4.3.

Figure 6 (left) shows the distribution of the number of bidders per auction for younger versus older users in both markets. On average, ‘Country 1’ (top) has more bidders per auction than ‘Country 2’ (bottom), and both markets have more bidders per auctions for the older users compared to the younger ones. The histograms in Figure 6 (right) reveal that video bidders tend to place higher bids than audio bidders. Notably, in contrast to Section 5.1, the ground truth value densities of the bidders are now unobserved.

Figure 5 (right) shows the profile likelihood, where the optimal discount $\delta_{OPT} = 0.9$, and the confidence interval is narrow due to the large sample size. Therefore, for the sake of brevity, we focus our analysis on δ_{OPT} . The dashed lines in Figure 6 show the estimated distributions, which closely align with the histograms (Figure 6, left).

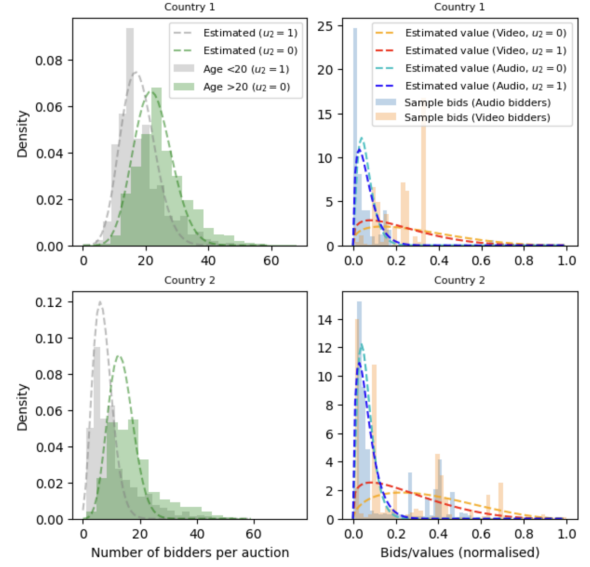


Figure 6: Two European markets, two age groups. Left: Distribution of the number of bidders in an auction. Right: Auction bids and the respective estimated value distributions.

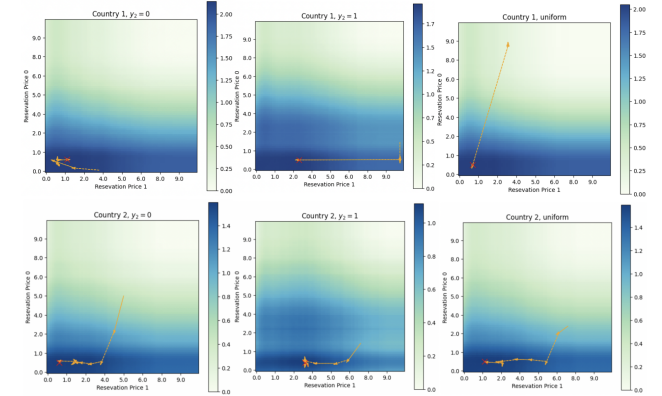


Figure 7: Revenue surfaces and optimal reservation prices in two markets (all prices are normalised). Left: users of type $y_2 = 0$. Centre: users of type $y_2 = 1$. Right: uniform (homogeneous) case.

From Figure 6 (right) we see that our model identifies significantly different value distributions for the different user types.

Figure 7 illustrates the resulting revenue surfaces and the gradient descent steps (orange arrows). The model identifies different optimal reservation prices for the various user types. Specifically, the optimal prices for older users ($y_2 = 0$) in ‘Country 1’ ($y_1 = 1$) are 0.75 for video and 0.5 for audio bidders (Figure 6, top left). For younger users ($y_2 = 1$) in the same country, the optimal prices are 2.88 for video bidders and 0.62 for audio bidders (Figure 6, top center and Table 2). Figure 6 (right) shows that the optimal prices in the homogeneous case are 0.75 for both video and audio bidders.

Table 2 shows that capturing heterogeneous bidders’ preferences results in higher auction revenue. For ‘Country 1’, we observe a +1.2% revenue increase when using personalised prices compared to MP. These revenue gains become even larger when compared with the approaches that do not take the strategic behaviour into

⁴These entire surfaces were constructed for the illustrative purposes only by running Algorithm 1 for all possible reservation prices within a given discrete range.

Country	Opt. Prices (vid/aud)		Revenue gains of P-MDN rel. to:			
	$y_2 = 0$	$y_2 = 1$	ZP	MP	ERM	SSM
$y_1 = 1$	0.75 / 0.5	2.88 / 0.62	+2.3%	+1.2%	+24.1%	+15.0%
$y_1 = 0$	1.25 / 0.75	2.50 / 0.50	+9.4%	+1.7%	+45.3%	+6.9%

Table 2: Optimal prices and the respective revenue gains of P-MDN relative to ZP, MP, ERM and SSM baselines.

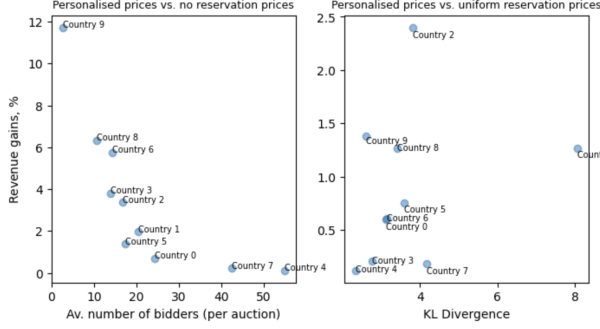


Figure 8: Relative revenue gains across ten markets. Left: Dependent on the number of bidders per auction. Right: Dependent on degree of heterogeneity of advertisers' preferences.

account: Specifically, we see +24.1% and +15% revenue increase relative to ERM and SSM respectively. The results for 'Country 2' are similar (see Table 2, last row).

5.3 Comparison Across Markets

We now illustrate how the revenue gains obtained by optimal prices are influenced by market competitiveness and the degree of heterogeneity of advertisers' preferences. In this section, we perform a similar analysis to Section 5.2, focusing on ten different markets.

Figure 8 (left) shows the relative revenue gains from using optimal personalised reservation prices compared to ZP across ten real markets (the respective revenue surfaces can be found in Figure 9 in the Appendix). On average, personalised prices increase auction revenue by +4%, with gains being consistently positive and substantial in some cases—reaching as high as +11.8% in 'Country 9'. This figure suggests that revenue gains may be influenced by market competitiveness. Specifically, as the number of bidders per auction increases, the respective gains tend to decrease. This is expected as with more bidders it becomes less likely that bidders will pay their reservation prices, thereby reducing the impact of those prices.

Figure 8 also illustrates the revenue gains from using personalising (heterogeneous) prices compared to using uniform ones. For all markets, the revenue gains from personalization are positive. We quantify the degree of heterogeneity in advertisers' preferences using the KL-divergence [24] of the estimated value distributions for different user types (i.e., the more dissimilar the distributions, the larger the KL-divergence). As shown in Figure 8 (right), revenue gains increase with higher KL-divergence. Intuitively, as advertisers' preferences become more diverse, it becomes easier to implement price differentiation and extract additional revenue.

5.4 Discussion

Our experiments suggest that our approach accurately estimates advertisers' unobservable values based on their observable bids. The parametric value model effectively captures the heterogeneity in

advertisers' preferences, allowing us to leverage this heterogeneity to boost auction revenue. Specifically, we found that personalised prices yield additional revenue gains compared to uniform ones.

Using a bidding dataset from real impression GSP auctions across ten markets, we demonstrated that personalising the prices consistently leads to positive revenue gains. The magnitude of these gains varies and may depend the level of market competitiveness (e.g., the average number of bidders per auction) and the degree of heterogeneity in advertisers' preferences.

6 Conclusion

This paper addresses the challenge of optimising revenue in impression GSP auctions by setting optimal personalised reservation prices. To this end, we derived the symmetric NE specifically for impression GSP auctions, and we used this NE to estimate the unobservable values of bidders based on their observable bids. To achieve this, we proposed a MDN architecture that incorporates the NE into its loss function, enabling us to infer advertisers' unobservable values directly from their bidding data. To tackle the parameter unidentifiability challenge that is particular for CPM auctions, we derived a two-stage estimation procedure combining maximum likelihood estimation with profile likelihood, and we developed an algorithm to compute user- and bidder-specific prices.

Our experiments demonstrated that our estimation method accurately captures the heterogeneity in advertisers' values. Using real bidding data from impression GSP auctions across ten markets, we showed that our approach can increase auction revenue by an average of 4%. These revenue gains are influenced by factors such as market competitiveness and the degree of heterogeneity in advertisers' preferences. Our findings suggest that implementing personalised prices in impression GSP can significantly boost revenue, particularly in markets with diverse advertiser preferences.

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A Appendix

A.1 Standard Definitions

An auction $\mathcal{A} = (g, p)$ is defined by a tuple of the *allocation* function $g : \mathbb{R}_{\geq 0}^N \rightarrow (\mathbb{N} \cup \{\infty\})^N$ and the *payment* function $p : \mathbb{R}_{\geq 0}^N \rightarrow \mathbb{R}_{\geq 0}^N$. Here, the allocation function $g(b)$ maps the bids b reported by bidders into the (deterministic) allocation $g(b) = a$, where $a = (a_1, \dots, a_N)$. The payment function $p(b)$ maps the bids profile b into the vector of payments $p = (p_1, \dots, p_N)$. We denote the bidder profile excluding bidder j 's bid as $b_{-j} = (b_1, \dots, b_{j-1}, b_{j+1}, \dots, b_N)$.

In the domain of impression GSP auctions, the allocation function can be defined as $g(b_1, \dots, b_N) = (a_1, \dots, a_N)$, where

$$a_j = \begin{cases} j & \text{if } j \leq K \\ \infty & \text{otherwise.} \end{cases} \quad (5)$$

Definition 1. A strategy profile $b^* = (b_1^*, \dots, b_N^*)$ is a **Nash equilibrium (NE)** of auction $\mathcal{A}$ if $\forall j = 1, \dots, N, \forall v_j, \forall \hat{b}_j$:

$$u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j) \geq u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j). \quad (6)$$

In words, the strategy profile b^* is a NE if for any bidder j there does not exist a beneficial deviation $\hat{b}_j$ from their strategy b_j^* .

A.2 Proofs

The following lemma demonstrates that with the bid profile defined in Equation (3) the bidders with larger values submit larger bids:

Lemma 1. *Bids profile b^* is value-ordered, i.e., $\forall i, j = 1, \dots, N$ $v_i \geq v_j$ implies $b_i^* \geq b_j^*$.*

PROOF. W.l.o.g., assume $v_1 \geq \dots \geq v_N$. We prove our statement by induction. First, observe that $b_{K+1}^* = \delta^K v_{K+1} \geq \dots \geq \delta^{N-1} v_N = b_N^*$, so that the bids profile $(b_{K+1}^*, \dots, b_N^*)$ is value-ordered. Now, consider $b_K^* = b_{K+1}^* + (\delta^{K-1} - \delta^K) v_K$. As $(\delta^{K-1} - \delta^K) v_K \geq 0$ this implies $b_K^* \geq b_{K+1}^*$. Consequently, $b_K^* \geq b_{K+j}^*$ for any $j = 1, \dots, N - K$. Thus, for all j , s.t., $v_K \geq v_j$ we also have $b_K^* \geq b_j^*$. Therefore $(b_K^*, \dots, b_N^*)$ is also value-ordered. This concludes the induction step. Q.E.D. $\square$

Theorem 1. *b^* is a Nash equilibrium of the impression GSP auction.*

PROOF. W.l.o.g., we let $v_1 \geq \dots \geq v_N$. From Lemma 1 we know that the bids b^* defined by Equation 3 are value-ordered, i.e., $b_1^* \geq \dots \geq b_N^*$. Therefore, bidder j gets allocated to slot j if $j \leq K$, bidders $j = K + 1, \dots, N$ are not allocated. Consider bidder j with bid $b_j^* = (\delta^{j-1} - \delta^j) v_j + b_{j+1}^*$. From Eq. 2 the utility of this bidder is

$$u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j) = v_j \delta^j - p_j = v_j \delta^j - b_{j+1}^*. \quad (7)$$

Let $\hat{b}_j$ be a deviating bid of bidder j . We now would like to illustrate that there does not exist a beneficial deviation for this bidder.

First, consider an upper-deviation $\hat{b}_j > b_j^*$. Assume $\hat{b}_j \geq b_{j-\ell}^*$, where $\ell \in \{1, 2, \dots, j-1\}$, s.t., bidder j now gets allocated to position $j - \ell$. In this case, $u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) = v_j \delta^{j-\ell} - b_{j+1}^*$. Therefore,

$$\begin{aligned} u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) - u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j) &= \\ (\delta^{j-\ell} - \delta^j) v_j - \underbrace{(b_{j-\ell}^* - b_{j+1}^*)}_{\geq b_{j-\ell+1}^* - b_{j+1}^*}. \end{aligned} \quad (8)$$

As the bids are value-ordered, we have:

$$\begin{aligned} b_{j-\ell}^* - b_{j+1}^* &= b_{j-\ell}^* - b_{j-\ell+1}^* + b_{j-\ell+1}^* - b_{j+1}^* = \sum_{k=0}^{\ell} (b_{j-\ell+k}^* - b_{j-\ell+k+1}^*) \geq \\ \sum_{k=0}^{\ell} (\delta^{j-\ell+k-1} - \delta^{j-\ell+k}) v_j &= (\delta^{j-\ell-1} - \delta^j) v_j. \end{aligned} \quad (9)$$

Consequently, $b_{j-\ell+1}^* - b_{j+1}^* \geq (\delta^{j-\ell} - \delta^j) v_j$. Comparing this with Eq. 8 we conclude that $u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) \leq u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j)$. Thus, $\hat{b}_j$ is not a beneficial deviation.

Now, consider a lower-deviation $\hat{b}_j < b_j^*$. Specifically, assume $\hat{b}_j < b_{j+\ell}^*$ where $\ell \in \{1, \dots, K - j\}$, s.t., j now get allocated to position $j + \ell$. The utility of bidder j from this deviation is

$$u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) = v_j \delta^{j+\ell} - b_{j+\ell+1}^*. \quad (10)$$

Consequently,

$$\begin{aligned} u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) - u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j) &= \\ (\delta^{j+\ell} - \delta^j) v_j - \underbrace{(b_{j+\ell+1}^* - b_{j+1}^*)}_{\leq b_{j+1}^* - b_{j+\ell}^*}. \end{aligned} \quad (11)$$

Now,

$$\begin{aligned} b_{j+1}^* - b_{j+\ell+1}^* &= \sum_{k=0}^{\ell-1} (b_{j+k+1}^* - b_{j+k+2}^*) = \sum_{k=0}^{\ell-1} (\delta^{j+k} - \delta^{j+k+1}) v_{j+k+1} \leq \\ \sum_{k=0}^{\ell-1} (\delta^{j+k} - \delta^{j+k+1}) v_{j+1} &= (\delta^j - \delta^{j+\ell}) v_{j+1} \leq (\delta^j - \delta^{j+\ell}) v_j. \end{aligned} \quad (12)$$

Comparing Equation 11 with Equation 12 we conclude that

$$u_j(g(\hat{b}_j, b_{-j}^*), p(\hat{b}_j, b_{-j}^*) | v_j) \leq u_j(g(b_j^*, b_{-j}^*), p(b_j^*, b_{-j}^*) | v_j).$$

Therefore, $\hat{b}_j$ is not a beneficial lower-deviation.

Finally, observe that for $j > K$ there does not exist a beneficial lower deviation as those bidders are not allocated anyway (and therefore, have zero utility for any lower deviation). They also do not have a beneficial upper deviation as in this case, they would have paid at least $b_K^* = v_{K+1} + (\delta^{K-1} - \delta^K) v_K \geq v_{K+1}$. But $b_K \geq v_j$ for all $j = K + 1, \dots, N$. Q.E.D. $\square$

Note that since the utility in the CPM domain differs from that in the CPC pricing domain [1], the respective symmetric equilibria in these two domains are also different. For comparison, we present the symmetric NE in GSP with CPC pricing as derived in [1, 7]:

$$b_j^{CPC} = \begin{cases} (1 - \delta) v_j + \delta b_{j+1}^{CPC} & j = 1, \dots, K \\ v_j & \text{otherwise.} \end{cases} \quad (13)$$

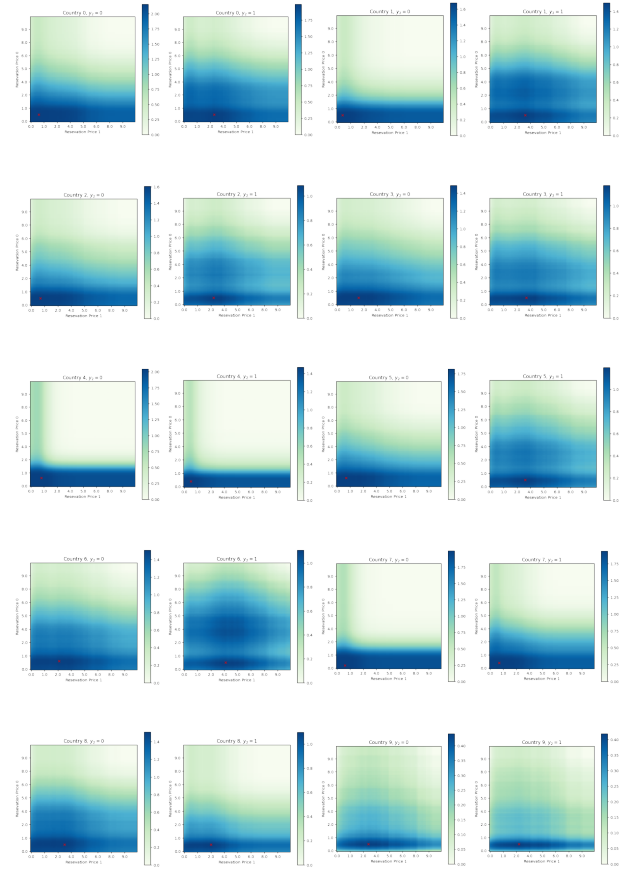


Figure 9: Revenue surfaces and the optimal prices for ten markets.

GenAI Usage Disclosure

There was no use of GenAI tools in any stage of the research.

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